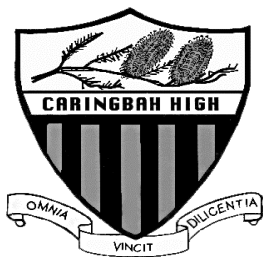


## *Caringbah High School*



### **2016** **Year 12 Trial HSC** **Examination**

# **Mathematics Extension 2**

#### **General Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen.
- Board-approved calculators may be used.
- A reference sheet with standard formulae and integrals is provided.
- In Questions 11–16, show relevant mathematical reasoning and/or calculations.

#### **Total marks – 100**

#### **Section I** Pages 3–6

**10 marks**

- Attempt Questions 1–10
- Allow about 15 minutes for this section

#### **Section II** Pages 7–13

**90 marks**

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section.

## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

---

1. Given that  $z = 5 + 2i$  and  $w = i - 3$ , what is the value of  $2\bar{w} - z$ ?

(A)  $-11 - 4i$  (B)  $-11$   
(C)  $-1 - 4i$  (D)  $-1$

2. The equation  $x^2 - y^2 - 5xy + 5 = 0$  defines  $y$  implicitly as a function of  $x$ .

What is the value of  $\frac{dy}{dx}$  at the point  $(1,1)$ ?

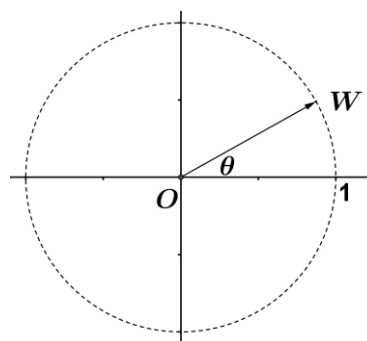
(A)  $-1$  (B)  $-\frac{3}{7}$   
(C)  $\frac{3}{7}$  (D)  $\frac{7}{3}$

3. What is the eccentricity of the conic described by the equation  $9x^2 - 4y^2 = 1$ ?

(A)  $\sqrt{\frac{5}{2}}$  (B)  $\sqrt{\frac{5}{3}}$   
(C)  $\frac{\sqrt{13}}{2}$  (D)  $\frac{\sqrt{13}}{3}$

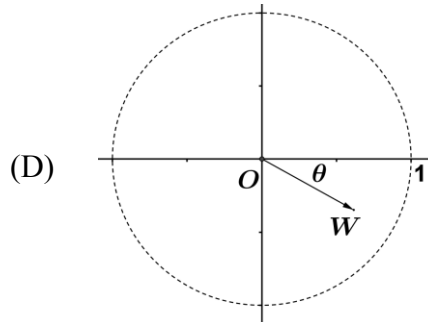
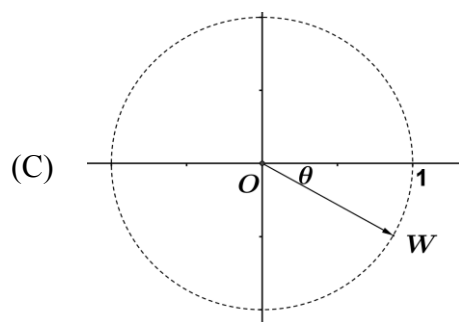
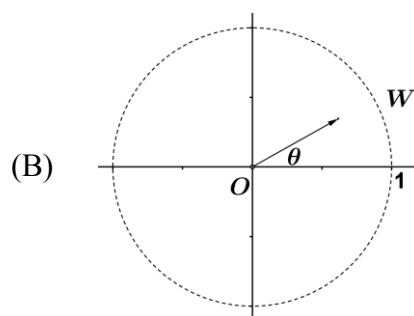
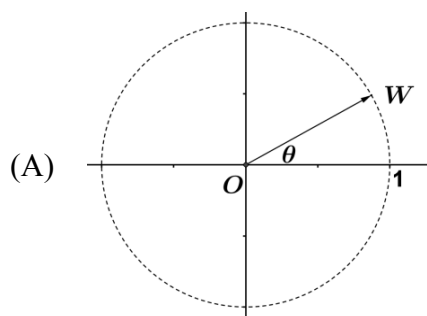
4.  $Z = x + iy$  is a complex number, represented on the Argand diagram as shown.

$$|Z| = 1.$$



Which of the following diagrams would represent

the complex number  $W = \frac{1}{\bar{Z}}$ ?



5. The equation  $4x^3 - 4x^2 - 15x + 18 = 0$  has a double root at  $x = \alpha$ .

The value of  $\alpha$  is

(A)  $\alpha = \frac{-3}{2}$

(B)  $\alpha = \frac{3}{2}$

(C)  $\alpha = \frac{5}{6}$

(D)  $\alpha = \frac{-5}{6}$

6. The conic described by the equation  $\frac{x^2}{169} + \frac{y^2}{25} = 1$  has directrices

(A)  $x = \pm 12$  (B)  $x = \pm \frac{169}{12}$

(C)  $y = \pm 12$  (D)  $y = \pm \frac{169}{12}$

7. If  $z = 2 - \sqrt{12}i$ , find the argument of  $z^5$ .

(A)  $\frac{-\pi}{3}$  (B)  $\frac{-5\pi}{6}$

(C)  $\frac{\pi}{3}$  (D)  $\frac{5\pi}{6}$

8. The complex number  $z$  satisfies  $\arg\left(\frac{z-2}{z+2i}\right) = \frac{-\pi}{2}$ .

Find the maximum value of  $|z|$ .

(A)  $\sqrt{2}$  (B)  $2\sqrt{2}$

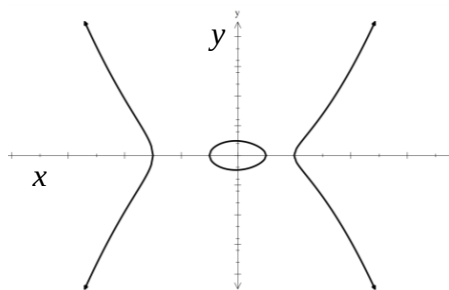
(C)  $2 - \sqrt{2}$  (D)  $2 + \sqrt{2}$

9. Which of the following is an equivalent expression for  $\int \frac{dx}{\sqrt{5-4x-x^2}}$  ?

(A)  $\sin^{-1}(x+2) + C$  (B)  $\sin^{-1}\left(\frac{x+2}{3}\right) + C$

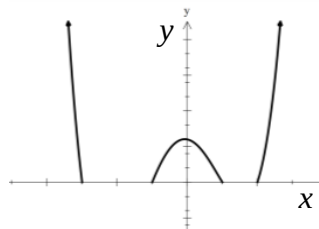
(C)  $\sin^{-1}(x-2) + C$  (D)  $\sin^{-1}\left(\frac{x-2}{3}\right) + C$

10.  $P(x)$  is a polynomial. The graph of  $y^2 = P(x)$  is shown below.

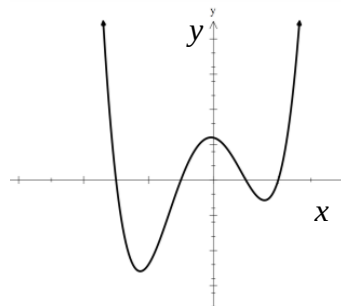


Which of the following graphs is the best representation of  $y = P(x)$ ?

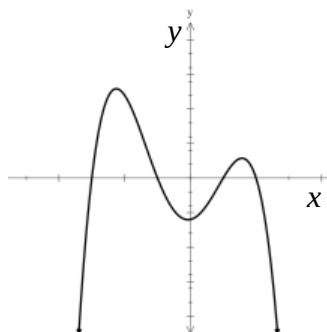
(A)



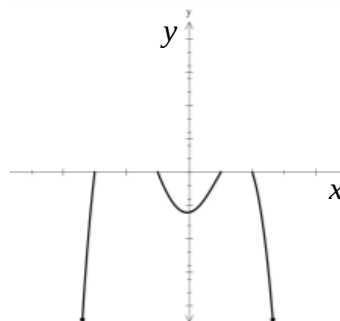
(B)



(C)



(D)



## Section II

90 marks

### Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

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| Question 11 (15 Marks)                                                                     | Marks |
|--------------------------------------------------------------------------------------------|-------|
| a) If $z = 1 - i$ , find $z^{-6}$ in the form $x + iy$ .                                   | 2     |
| b) If $z = x + iy$ , shade on the Argand diagram the region defined by $z\bar{z} \leq 4$ . | 2     |
| c) Evaluate $\int_{-1}^1 \frac{dx}{x^2 + 2x + 5}$                                          | 3     |
| d) Show that $\int_e^{e^4} \ln x \, dx = 3e^4$                                             | 3     |
| e) i) Show that $(z - 2i)$ is a factor of $P(z) = z^4 + z^3 + z^2 + 4z - 12$ .             | 1     |
| ii) Hence, find all zeros of $P(z)$ .                                                      | 1     |
| f) Let $A = 3 + 4i$ and $B = 9 + 4i$ be two points on the Argand diagram.                  |       |
| i) Sketch the locus defined by $ z - A  = 5$                                               | 1     |
| ii) Draw a clear sketch of the curve defined by $ z - A  +  z - B  = 12$                   | 2     |

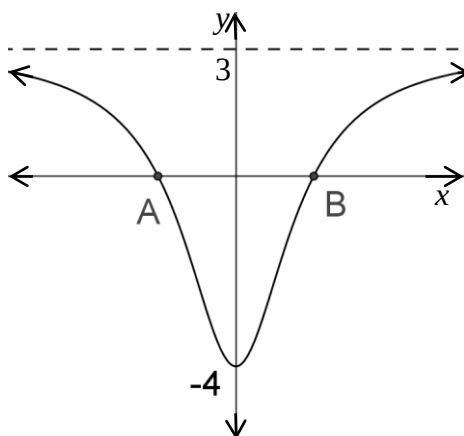
**End of Question 11**

**Question 12 (15 Marks)****Marks**

- a) i) Express  $3\sin\theta + 4\cos\theta$  in the form  $r\sin(\theta + \alpha)$ , where  $r$  and  $\alpha$  are constants and  $\alpha$  is in radians correct to 3 decimal places. 2
- ii) Hence, or otherwise, show that a particle whose displacement  $x$  metres, after  $t$  seconds, given by  $x = 8\cos^2 t + 6\sin t \cos t - 4$  is moving in simple harmonic motion. 2

You may assume that  $2\cos A \sin B = \sin(A + B) - \sin(A - B)$

- b) The graph of  $y = f(x)$  is shown.



On separate diagrams, show the following graphs, clearly indicating important features.

- i)  $y = \frac{1}{f(x)}$  2
- ii)  $y = [f(x)]^2$  2
- iii)  $y = \log_e(f(x))$  2

**Question 12 continues on page 9**

Question 12 (continued)

- c) The complex number  $z = x + iy$ , with  $x$  and  $y$  real, satisfies  $|z - i| = \operatorname{Im}(z)$ .
- i) Show that the locus of the point  $P$ , representing  $z$ , has the **2**  
Cartesian equation  $y = \frac{x^2 + 1}{2}$
- ii) By finding the gradients of the tangents to this curve that pass through **3**  
the origin, state the set of possible values for the principal argument of  $z$ .

**End of Question 12**

**Question 13 (15 Marks)****Marks**

- a) The base of a solid is the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ . 4

Cross-sections perpendicular to the  $x$ -axis are right isosceles triangles with one of the equal sides in the base of the solid.

Show that the volume of the solid is  $\frac{8ab^2}{3}$  units<sup>3</sup>.

- b) Sketch the curve  $y = x \ln(x)$ , showing any turning points. 2

- c) i) If  $z$  is a complex number defined by  $z = \cos\theta + i\sin\theta$ , 1  
show that  $\frac{dz}{d\theta} = iz$

- ii) By integrating  $\frac{dz}{d\theta} = iz$  with respect to  $z$  and  $\theta$ , 2  
show that  $z$  can be written in the form  $z = e^{i\theta}$ .

- d) i) Evaluate  $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x + \sin x}$  using the substitution  $t = \tan \frac{x}{2}$ . 3

- ii) Hence evaluate  $\int_0^{\frac{\pi}{2}} \frac{x dx}{1 + \cos x + \sin x}$  3  
using the substitution  $u = \frac{\pi}{2} - x$ .

**End of Question 13**

**Question 14 (15 Marks)**

**Marks**

a) Sketch on the Argand diagram, the locus defined by  $\arg(z - 1 + i) = \frac{\pi}{4}$  **1**

b) Two points,  $P\left(cp, \frac{c}{p}\right)$  and  $Q\left(cq, \frac{c}{q}\right)$ , lie on the rectangular hyperbola  $xy = c^2$ .

i) Show that the equation of the tangent at  $P$  is  $x + p^2y = 2cp$ . **2**

ii) The tangents at  $P$  and  $Q$  meet in  $T$ . Find the coordinates of  $T$  in terms of  $c$ ,  $p$  and  $q$ . **2**

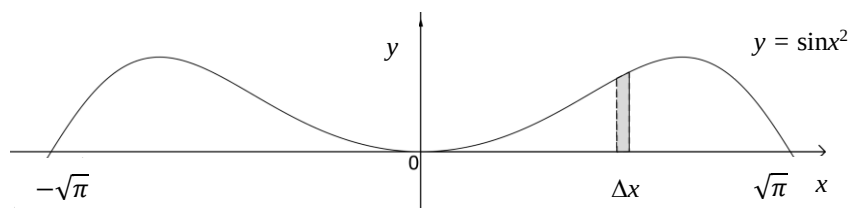
iii) The point  $T$  lies on another hyperbola  $xy = k^2$ , for all positions of  $P$  and  $Q$ . **1**

Show that  $\frac{pq}{(p+q)^2} = \frac{k^2}{4c^2}$

c) i) Let  $I_n = \int_1^e x(\ln x)^n dx$ . Show that  $I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}$ ,  $n = 1, 2, 3, \dots$  **3**

ii) Hence evaluate  $\int_1^e x(\ln x)^2 dx$  **2**

d) The area bounded by the curve  $y = \sin x^2$  and the  $x$ -axis, in the domain  $-\sqrt{\pi} \leq x \leq \sqrt{\pi}$ , is rotated about the  $y$ -axis. **4**



By using the method of cylindrical shells, find the volume of the solid formed.

**End of Question 14**

**Question 15 (15 Marks)****Marks**

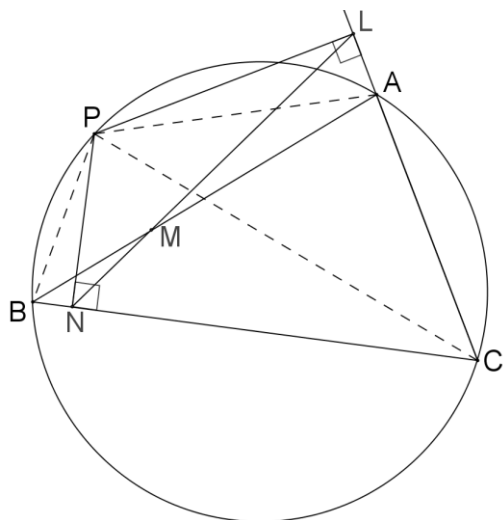
- a)  $\alpha, \beta$  and  $\gamma$  are roots of the cubic equation  $x^3 + mx + n = 0$ .
- i) Find the value of  $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$  in terms of  $m$  and  $n$ . 2
- ii) Find the value of  $\alpha^3 + \beta^3 + \gamma^3$  in terms of  $m$  and  $n$ . 2
- iii) Determine the cubic equation with roots  $\alpha^2, \beta^2$  and  $\gamma^2$ . 2
- b) For the ellipse  $E : 4x^2 + 9y^2 = 36$ .
- i) Sketch the ellipse  $E$  indicating the position of its foci  $S$  and  $S'$ , and draw in its directrices. 3
- ii) Show that the point  $P(3\cos\theta, 2\sin\theta)$  lies on  $E$ . 1
- iii) Derive the equation of the tangent to  $E$  at the point  $P$ . 2
- iv) Find the coordinates of the point  $Q$  where the tangent cuts the major axis. 1
- v) The equation of the normal at  $P$  is  $\frac{3x}{\cos\theta} - \frac{2y}{\sin\theta} = 5$  and it cuts the major axis at  $R$ . A line parallel to the  $y$ -axis through  $P$  cuts the  $x$ -axis at  $T$ . Show that  $OQ \times RT$  is constant for all positions of  $P$ . 2

**End of Question 15**

**Question 16 (15 Marks)**

**Marks**

a)



$ABC$  is an acute-angled triangle inscribed in a circle.  $P$  is a point on the minor arc  $AB$  of the circle.  $PL$  and  $PN$  are the perpendiculars from  $P$  to  $CA$  produce and  $CB$  respectively.  $LN$  cuts  $AB$  at  $M$ .

i) Copy the diagram in to your answer booklet. ( $\frac{1}{3}$  page)

ii) Explain why  $PNCL$  is a cyclic quadrilateral.

**1**

iii) Hence show  $PBNM$  is also a cyclic quadrilateral.

**3**

iv) Hence show that  $PM$  is perpendicular to  $AB$ .

**2**

b) i) Write expressions for  $\cos \frac{2\pi}{n}$  and  $\sin \frac{2\pi}{n}$  in terms of  $\cos \frac{\pi}{n}$  and  $\sin \frac{\pi}{n}$

**2**

ii) By using De Moivre's theorem, show that

**3**

$$\left(1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}\right)^n = -2^n \left(\cos \frac{\pi}{n}\right)^n$$

c) i) If  $a$  and  $b$  are both positive numbers, prove that  $a + b \geq 2\sqrt{ab}$ .

**1**

Hence, or otherwise, prove that when  $a$ ,  $b$  and  $c$  are all positive numbers and  $a + b + c = 1$ , then

**3**

$$\left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right) \geq 8$$

**End of Examination**

1) A 2) B 3) C 4) A 5) B

6) B 7) C 8) B 9) B 10) B

### Question 11

a)  $z = 1 - i = \sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right)$

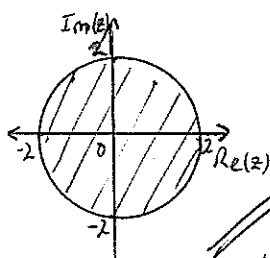
$z^{-6} = (\sqrt{2})^{-6} \operatorname{cis}\left(\frac{6\pi}{4}\right)$

$= \frac{1}{8} \operatorname{cis}\left(-\frac{\pi}{2}\right)$

$= -\frac{i}{8}$

b)  $(x+iy)(x-iy) = x^2 + y^2$

$z \cdot \bar{z} \leq 4$



c)  $\int_{-1}^1 \frac{dx}{x^2+2x+5} = \int_{-1}^1 \frac{dx}{(x+1)^2+4}$

$= \frac{1}{2} \left[ \tan^{-1}\left(\frac{x+1}{2}\right) \right]_{-1}^1$

$= \frac{1}{2} \left[ \tan^{-1}1 - \tan^{-1}0 \right]$

$= \frac{\pi}{8}$

d)  $\int_e^{e^4} \ln x dx$   $u = \ln x$   $dv = dx$

$du = \frac{1}{x} dx$   $v = x$

$= x \ln x \Big|_e^{e^4} - \int_e^{e^4} x \times \frac{1}{x} dx$

$= [e^4 \ln e^4 - e \ln e] - [x]_e^{e^4}$

$= 4e^4 - e - e^4 + e$

$= 3e^4$

e)  $P(2i) = (2i)^4 + (2i)^3 + (2i)^2 + 4 \times 2i - 12$

$= 16 - 8i - 4 + 8i - 12$

$= 0$

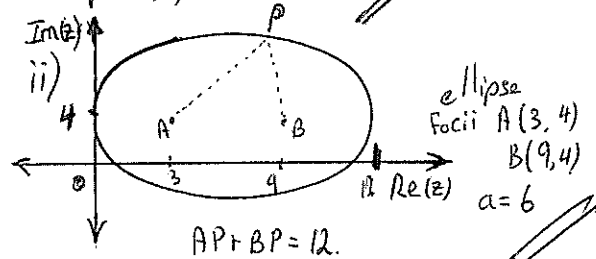
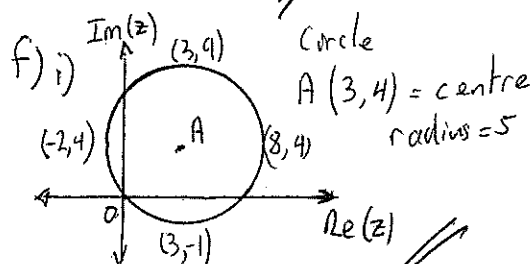
$\therefore (z-2i)$  is a factor of  $P(z)$

ii)  $P(z)$  has real coefficients  $\therefore (z+2i)$  is a factor

$\therefore (z+2i)(z-2i) = (z^2+4)$  is a factor.

$z^4 + z^3 + z^2 + 4z - 12 = (z^2+4)(z^2+z-3)$

roots are  $\pm 2i, \frac{-1 \pm \sqrt{1+12}}{2} = \frac{-1 \pm \sqrt{13}}{2}$



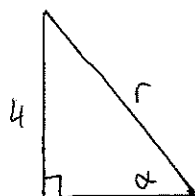
### Question 12

a)  $3 \sin \theta + 4 \cos \theta$

$= r \sin \theta \cos \alpha + r \cos \theta \sin \alpha$

$r \cos \alpha = 3$

$r \sin \alpha = 4$



$r = 5$   $\tan \alpha = \frac{4}{3}$

$\alpha \approx 0.927$

$3 \sin \theta + 4 \cos \theta = 5 \sin(\theta + 0.927)$

ii)  $3$

$x = 8 \cos^2 t + 6 \sin t \cos t - 4$

$= 8 \cos^2 t + 3 \sin 2t - 4$

$\dot{x} = 16 \cos t \sin t + 6 \cos 2t$

$= -8 \sin 2t + 6 \cos 2t$

$\ddot{x} = -16 \cos 2t - 12 \sin 2t$

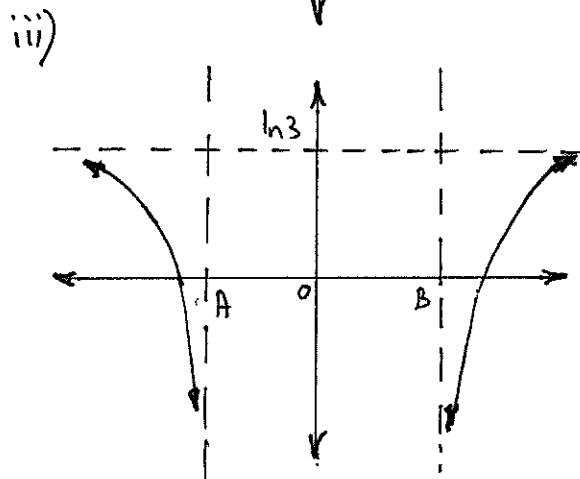
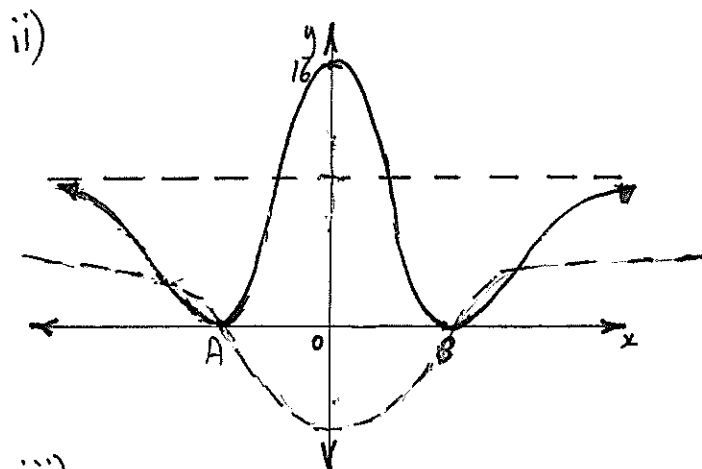
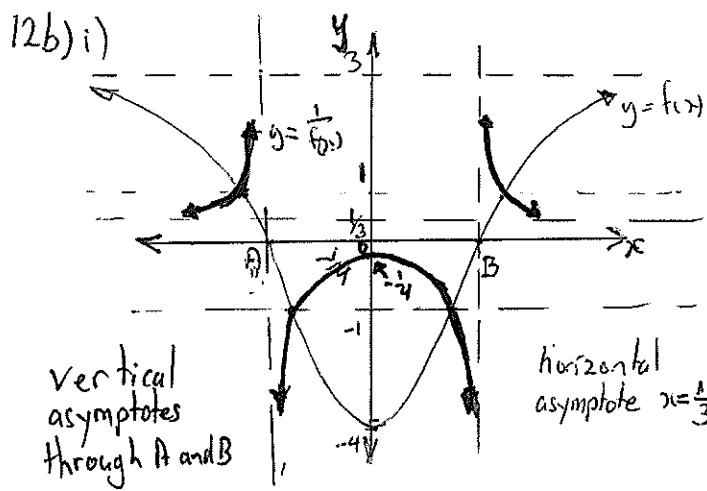
$= -16(2 \cos^2 t - 1) - 24 \sin t \cos t$

$= -32 \cos^2 t - 24 \sin t \cos t + 16$

$= -4(8 \cos^2 t + 6 \sin t \cos t - 4)$

$= -4x$

$\therefore$  SHM,  $n=2$ ,  $T = \pi$



c) i)  $|z - i| = \text{Im } z$

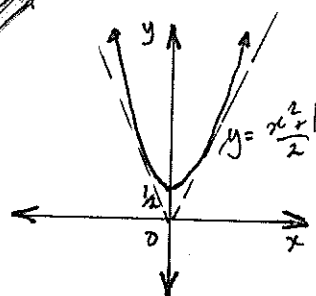
$$\sqrt{(x^2 + (y-1)^2)} = y$$

$$x^2 + y^2 - 2y + 1 = y^2$$

$$x^2 - 2y + 1 = 0$$

$$2y = x^2 + 1$$

$$y = \frac{x^2 + 1}{2}$$



ii)  $y = \frac{x^2}{2} + \frac{1}{2}$

$\frac{dy}{dx} = x$  If it passes through 0, then gradient =  $\frac{y}{x}$

$$\frac{y}{x} = x$$

$$y = x^2$$

$$\frac{x^2 + 1}{2} = x^2$$

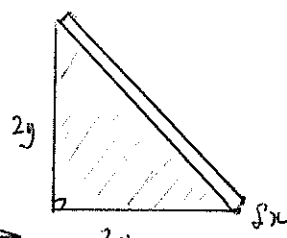
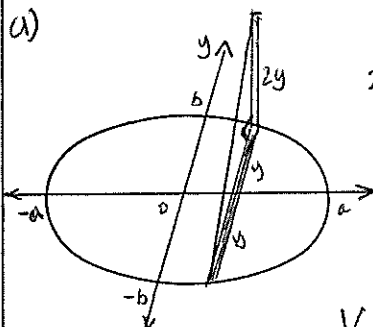
$$x^2 = 1 \therefore x = \pm 1, y = 1$$

ie tangents inclined at  $\frac{\pi}{4}$  and  $\frac{3\pi}{4}$

$$\therefore \frac{\pi}{4} \leq \arg(z) \leq \frac{3\pi}{4}$$

### Question 13

a)



$$\delta V = \frac{1}{2} \times (2y)^2 \delta x$$

$$V = 2 \int_0^a 2y^2 dx$$

$$y^2 = b^2 \left(1 - \frac{x^2}{a^2}\right)$$

$$= \frac{b^2}{a^2} (a^2 - x^2)$$

$$V = \frac{4b^2}{a^2} \int_0^a (a^2 - x^2) dx$$

$$= \frac{4b^2}{a^2} \left[ a^2 x - \frac{x^3}{3} \right]_0^a$$

$$= \frac{4b^2}{a^2} \left[ \left( a^3 - \frac{a^3}{3} \right) - 0 \right]$$

$$= \frac{4b^2}{a^2} \times \frac{2a^3}{3}$$

$$= \frac{8ab^2}{3}$$

b)  $y = x \ln x \Rightarrow x > 0$

$$y' = x \times \frac{1}{x} + \ln x$$

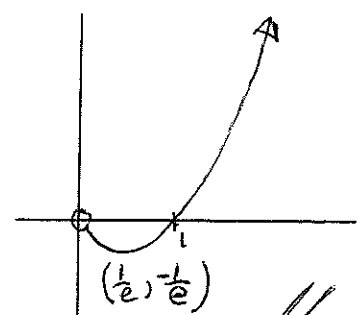
$$= 1 + \ln x$$

$$y' = 0 \ln x = -1$$

$$x = e^{-1}$$

$$= \frac{1}{e}$$

$$y = -\frac{1}{e}$$



$$y'' = \frac{1}{x} \therefore \text{min TP at } \left( \frac{1}{e}, -\frac{1}{e} \right)$$

$$\text{as } x > 0 \Rightarrow y'' > 0$$

ie always concave up.

13 c) i)  $z = \cos \theta + i \sin \theta$

$$\frac{dz}{d\theta} = -\sin \theta + i \cos \theta$$

$$i z = i (\cos \theta + i \sin \theta) \\ = i \cos \theta - \sin \theta \\ = \frac{dz}{d\theta}$$

ii)  $\frac{dz}{d\theta} = i z$

$$\int \frac{dz}{z} = \int i d\theta$$

$$\ln z = i\theta + c$$

$$\theta = 0 \quad z = \cos 0 + i \sin 0 = 1$$

$$\text{so } \ln 1 = 0 + c \quad \therefore c = 0$$

$$\ln z = i\theta \\ \underline{z = e^{i\theta}} \quad \text{[A useful extension to complex numbers!]}$$

d) i)

$$\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cos x + \sin x}$$

$$x = \tan^{-1} t \\ \sin x = \frac{2t}{1+t^2} \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$x = 2 \tan^{-1} t$$

$$dx = \frac{2dt}{1+t^2}$$

$$x=0 \quad t=0 \\ x=\frac{\pi}{2} \quad t=1$$

$$\int_0^1 \frac{2dt}{1+t^2 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}} \times \left( \frac{1+t^2}{1+t^2} \right)$$

$$\int_0^1 \frac{2dt}{1+t^2 + 1-t^2 + 2t}$$

$$\int_0^1 \frac{2dt}{2+2t}$$

$$\int_0^1 \frac{dt}{1+t} = \left[ \ln |1+t| \right]_0^1$$

$$= \ln 2 - \ln 1$$

$$= \ln 2$$

ii)  $\int_0^{\frac{\pi}{2}} \frac{x dx}{1 + \cos x + \sin x} = \int_{\frac{\pi}{2}}^0 \frac{(\frac{\pi}{2} - u) (-du)}{1 + \cos(\frac{\pi}{2} - u) + \sin(\frac{\pi}{2} - u)}$

$$u = \frac{\pi}{2} - x$$

$$du = -dx$$

$$x=0 \quad u = \frac{\pi}{2}$$

$$x = \frac{\pi}{2} \quad u = 0$$

$$= \int_{\frac{\pi}{2}}^0 \frac{-\frac{\pi}{2} du}{1 + \sin u + \cos u} - \int_{\frac{\pi}{2}}^0 \frac{-u du}{1 + \cos u + \sin u}$$

$$= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \frac{du}{1 + \cos u + \sin u} - \int_0^{\frac{\pi}{2}} \frac{u}{1 + \cos u + \sin u}$$

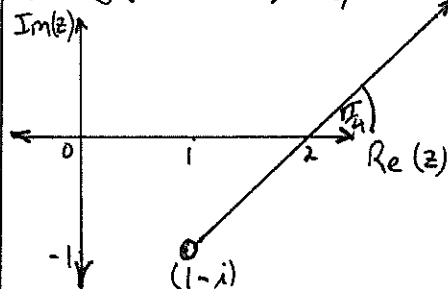
$$\text{let } u = x$$

$$2I = \frac{\pi}{2} \times \ln 2 \quad \text{[from i)]}$$

$$I = \frac{\pi}{4} \ln 2$$

## Question 14

a)  $\arg(z - (1-i)) = \frac{\pi}{4}$



b) i)  $y = \frac{c^2}{x} \quad \frac{dy}{dx} = -\frac{c^2}{x^2}$

$$\text{at } x = cp \quad m = \frac{-c^2}{c^2 p^2} = -\frac{1}{p^2}$$

Eqn of tangent

$$(y - \frac{c}{p}) = -\frac{1}{p^2} (x - cp)$$

$$p^2 y - cp = -x + cp$$

$$x + p^2 y = 2cp$$

ii)  $x + p^2 y = 2cp$  Eqs of tangents

$$x + q^2 y = 2cq$$

$$y(p^2 - q^2) = 2c(p - q)$$

$$y(p - q)(p + q) = 2c(p - q)$$

$$\text{As } (p - q) \neq 0$$

$$y(p + q) = 2c$$

$$y = \frac{2c}{p+q}$$

$$x = 2cp - p^2 y$$

$$= 2cp - \frac{2cp^2}{p+q}$$

$$= \frac{2cp^2 + 2cpq}{p+q} - \frac{2cp^2}{p+q}$$

$$= \frac{2cpq}{p+q} \quad \text{coord of } T \left( \frac{2cpq}{p+q}, \frac{2c}{p+q} \right)$$

iii)  $xy = k^2 \Rightarrow \frac{2cpq}{(p+q)} \times \frac{2c}{(p+q)} = k^2$

$$\frac{4c^2 pq}{(p+q)^2} = k^2$$

$$\frac{pq}{(p+q)^2} = \frac{k^2}{4c^2}$$

14c) i)  $I_n = \int_1^e x (\ln x)^n dx$   $u = (\ln x)$   $dv = x dx$   
 $du = n(\ln x)^{n-1} \cdot \frac{1}{x} dx$   $v = \frac{x^2}{2}$

$$I_n = \frac{x^2}{2} (\ln x)^n \Big|_1^e - \int_1^e \frac{n x^2}{2} (\ln x)^{n-1} \cdot \frac{1}{x} dx$$

$$= \left[ \frac{e^2}{2} (\ln e)^n - \frac{1}{2} (\ln 1)^n \right] - \frac{n}{2} \int_1^e x (\ln x)^{n-1} dx$$

$$\ln e = 1 \quad \ln 1 = 0$$

$$= \frac{e^2}{2} - \frac{n}{2} I_{n-1}$$

ii)  $n=2$   
 $I_2 = \frac{e^2}{2} - I_1$

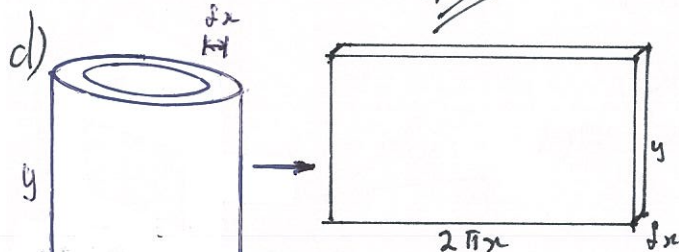
$n=1$   
 $I_1 = \frac{e^2}{2} - \frac{1}{2} I_0$

$$I_0 = \int_1^e x (\ln x)^0 dx = \left[ \frac{1}{2} x^2 \right]_1^e$$

$$= \frac{e^2}{2} - \frac{1}{2}$$

$$\therefore I_2 = \frac{e^2}{2} - \left[ \frac{e^2}{2} - \frac{1}{2} \left( \frac{e^2}{2} - \frac{1}{2} \right) \right]$$

$$= \frac{e^2}{4} - \frac{1}{4} \text{ or } \frac{e^2-1}{4}$$

d) 

$$r = 2\pi x y \int x$$

$$V = 2\pi \int_0^{\sqrt{\pi}} x y dx$$

$$= 2\pi \int_0^{\sqrt{\pi}} x \sinh x^2 dx$$

let  $u = x^2$   $du = 2x dx$

$x=0$   $u=0$

$x=\sqrt{\pi}$   $u=\pi$

$$V = \pi \int_0^{\pi} \sin u du$$

$$= \pi [-\cos u]_0^{\pi}$$

$$= \pi [1 + 1]$$

$$= 2\pi u^3$$

### Question 15

i)  $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \alpha\gamma}{\alpha\beta\gamma}$

$$x^3 + 0x^2 + mx + n = 0$$

$$-\frac{b}{a} = 0, \quad \frac{c}{a} = m = -\frac{d}{a} = -n$$

$$\text{So } \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = -\frac{m}{n}$$

ii)  $\alpha^3 = -m\alpha - n$

$$\beta^3 = -m\beta - n$$

$$\gamma^3 = -m\gamma - n$$

$$\alpha^3 + \beta^3 + \gamma^3 = -m(\alpha + \beta + \gamma) - 3n$$

$$\alpha + \beta + \gamma = 0$$

$$= -3n$$

iii) Sub  $x = \sqrt{x}$

$$(\sqrt{x})^3 + m\sqrt{x} + n = 0$$

$$x\sqrt{x} + m\sqrt{x} = -n$$

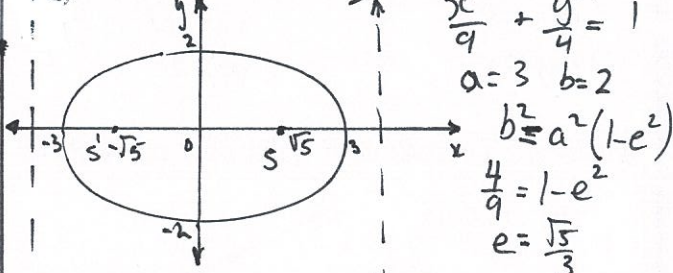
$$\sqrt{x}(x+m) = -n$$

squaring both sides

$$x(x+m)^2 = n^2$$

$$x^3 + 2mx^2 + m^2x - n^2 = 0$$

b)



$$x = -\frac{9}{\sqrt{5}}$$

$$y = \frac{9}{\sqrt{5}}$$

ii)  $\frac{(3 \cos \theta)^2}{9} + \frac{(2 \sin \theta)^2}{4} = \frac{9 \cos^2 \theta}{9} + \frac{4 \sin^2 \theta}{4}$

$$= \cos^2 \theta + \sin^2 \theta$$

$$= 1$$

$\therefore P$  lies on  $E$ .

iii)  $4x^2 + 9y^2 = 36$

$$8x + 18y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{8x}{18y}$$

$$= -\frac{4x}{9y}$$

at  $E$   $m = \frac{-4(3 \cos \theta)}{9(2 \sin \theta)}$

$$m = -\frac{2}{3} \frac{\cos \theta}{\sin \theta}$$

Eqn of tangent

$$(y - 2 \sin \theta) = \frac{-2 \cos \theta}{3 \sin \theta} (x - 3 \cos \theta)$$

$$3 \sin \theta y - 6 \sin^2 \theta = -2x \cos \theta + 6 \cos^2 \theta$$

$$2 \cos \theta x + 3 \sin \theta y = 6(\sin^2 \theta + \cos^2 \theta)$$

$$= 6$$

$$\frac{\cos \theta x}{3} + \frac{\sin \theta y}{2} = 1$$

15b i)  $y=0$   $2 \cos \theta x = 6$

$$x = \frac{3}{\cos \theta}$$

$$OQ = \frac{3}{\cos \theta}$$

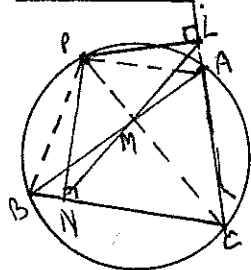
$$OT = 3 \cos \theta$$

$$OR = \frac{5 \cos \theta}{3}$$

$$\begin{aligned} RT &= OT - OR \\ &= 3 \cos \theta - \frac{5 \cos \theta}{3} \\ &= \frac{4 \cos \theta}{3} \end{aligned}$$

$$\begin{aligned} OQ \times RT &= \frac{3}{\cos \theta} \times \frac{4 \cos \theta}{3} \\ &= 4 \text{ ie constant for all positions of } P. \end{aligned}$$

Question 16



ii)  $\angle PLC = \angle PNC = 90^\circ$  (perpendicular)  
 $\therefore$  opp angles of quadrilateral are supplementary.

$\therefore$  PNC is a cyclic quad.

iii) As PNC are concyclic (from ii):  
 $\angle PNL = \angle PCL$  angles at circumference standing on chord PL

APBC are concyclic so  
 $\angle PBA = \angle PCA$  angles at circum. standing on chord PA

$\angle PCL$  is also  $\angle PCA$

so  $\angle PNL = \angle PCL = \angle PCA = \angle PBA$

or  $\angle PNL = \angle PBA$

$\angle PNL$  can be named  $\angle PNM$

$\angle PBA$  can be named  $\angle PBM$

so  $\angle PNM = \angle PBM$

$\therefore$  PBNM is a cyclic quad  
 equal angles at circumference on chord PM.

ii)  $\angle BNP$  and  $\angle BMP$  stand on chord BP in circle PBNM (from ii).

$\therefore \angle BNP = \angle BMP$

as  $PN \perp BC$  (given).

$\angle PMB = 90^\circ$

ie  $PM \perp AB$ .

b) i)  $\cos \frac{2\pi}{n} = \cos^2 \frac{\pi}{n} - \sin^2 \frac{\pi}{n}$

$\sin \frac{2\pi}{n} = 2 \sin \frac{\pi}{n} \cos \frac{\pi}{n}$

b ii)

$$1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n} = \left( \cos^2 \frac{\pi}{n} + i \sin^2 \frac{\pi}{n} \right) + \left( \cos^2 \frac{\pi}{n} - \sin^2 \frac{\pi}{n} \right) + (2i \sin \frac{\pi}{n} \cos \frac{\pi}{n})$$

$$= 2 \cos^2 \frac{\pi}{n} + 2i \sin \frac{\pi}{n} \cos \frac{\pi}{n}$$

$$= 2 \cos \frac{\pi}{n} \left( \cos \frac{\pi}{n} + i \sin \frac{\pi}{n} \right)$$

$$\left( 1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n} \right)^n = 2^n \left( \cos \frac{\pi}{n} \right)^n \left( \cos \frac{\pi}{n} + i \sin \frac{\pi}{n} \right)^n$$

$$= 2^n \left( \cos \frac{\pi}{n} \right)^n \left( \cos \pi + i \sin \pi \right)$$

$$e^{i\pi} = -1$$

$$= -2^n \left( \cos \frac{\pi}{n} \right)^n$$

c) i)  $(a-b)^2 \geq 0$

$$(a-b)^2 = a^2 + b^2 - 2ab \geq 0$$

$$(a+b)^2 - 4ab \geq 0$$

$$(a+b)^2 \geq 4ab$$

$$a+b \geq 2\sqrt{ab}$$

(positive case as a, b are both positive)

ii)

$$\left( \frac{1}{a} - 1 \right) = \frac{1-a}{a} \quad \text{If } a+b+c=1$$

$$a = 1 - b - c$$

$$= \frac{1 - (1 - b - c)}{a}$$

$$= \frac{b+c}{a}$$

$$\left( \frac{1}{a} - 1 \right) \left( \frac{1}{b} - 1 \right) \left( \frac{1}{c} - 1 \right) = \frac{(b+c)}{a} \frac{(a+c)}{b} \frac{(a+b)}{c}$$

from i)  $a+b \geq 2\sqrt{ab}$

$$\geq \frac{2\sqrt{bc}}{a} \times \frac{2\sqrt{ac}}{b} \times \frac{2\sqrt{ab}}{c}$$

$$\geq \frac{8\sqrt{a^2 b^2 c^2}}{abc}$$

$$\geq \frac{8abc}{abc}$$

$$\geq 8$$